NeatHack2026-Jevelopers: token usage, caching and API-equivalent cost

Snapshot date: 2026-10-12 · **Timezone for time-of-day analysis:** America/New_York · **Mode:** read-only, local logs only · **Prices:** list API rates as of 2026-10-12

Scope: NeatHack2026-Jevelopers on this Mac, October 10-12, 2026, through October 12 at 2:31:50 p.m. Eastern (inclusive). The accounting chat is excluded. 192 project session files contain 189 sessions with usage and 16,189 nonzero calls. Other projects and remote machines are excluded. Credential files and prompt text were not captured.

Executive summary

Total recorded usage is 3,859,621,041 tokens: 70,168,391 fresh input, 3,773,394,304 cached input, and 16,058,346 output. Output includes 8,592,119 reasoning tokens. Prompt cache hit rate is 98.2%. At verified standard API list rates, including per-call long-context premiums, the estimate is $3,320.68; the same calls with all input uncached would be $28,437.55. Actual subscription fees and billed charges are unknown.

Measure	Value
Model calls compiled	16,189
Sessions / threads	189
Typed prompts recovered	0
Active days	3
Total tokens (all four classes), de-duplicated	3,859,621,041 (3.86B)
Project-selected SQLite counter snapshot (different time; see validation)	3.69B
Output tokens (reasoning share)	16.06M (53.5%)
Cache hit rate, all tools	98.2%
API-equivalent cost at list rates	$3,320.68
Cost if nothing had been cached	$28,437.55
Saved by prompt caching	$25,116.87 (88.3%)
Actual subscription fees	Unknown
Account allocation of API-equivalent usage	Unknown
Account allocation of API-equivalent usage	Unknown
Peak month	2026-10: 3.86B tokens, $3,320.68 API-equivalent

Confidence definitions. *High*: read directly from per-call usage records the tool wrote at request time, de-duplicated, and cross-checked against a second counter. *Medium*: derived from those records through a documented rule with a stated assumption (pricing rows, replay filter, account rules). *Low*: an estimate that depends on an unpublished price or on a tool-maintained aggregate that could not be reconciled call by call.

1. Coverage and evidence rules

Three classes of evidence are used and kept apart throughout:

1. **Per-call usage records (class A).** Project-selected Codex rollout files; each nonzero call uses last_token_usage from a token_count event. All token totals come from these records. 2. **Tool-maintained aggregates (class B).** A read-only, project-selected snapshot of Codex threads.tokens_used. Differences caused by the reset counter and later activity are explained in validation. 3. **Derived estimates (class C).** API-equivalent cost, cache savings, subscription comparisons and account attribution. These multiply class A tokens by published list prices in `pricing.json` and apply the rules in `accounts.json`; models without a published rate are priced at a stated sibling.

Source inventory

Tool	Host	Files	Size	First	Last	Calls	Tokens
Codex	local	192	1.20 GB	2026-10-10	2026-10-12	16,189	3.86B

Hosts: **local** = This Mac: project-selected Codex Desktop/CLI rollout files.

Scan provenance

Host	Tool	Root scanned	Calls found	Prompts	Tool's thread count	Tool's token total
local	Codex	`.codex (project-selected rollout files only)`	16,189	0	192	3.69B

Scan roots are shown by their last path component only; the full paths stay on the machine unless the manifest sets `package_paths: true`.

Scan statistics: **local** 192 files, 1.2 GB, 167,090 lines, 0 JSON errors, 0 replayed Codex token events skipped, 612 duplicates skipped, 2.6s.

Sources examined and excluded

- This accounting conversation is excluded.
- Other projects, other machines, remote hosts, cloud-only conversations and calls missing from local logs are excluded.
- Prompt text, credential files and raw-log archives were not captured.

2. Method

2.1 Parsing rules

- **Claude Code.** Every `assistant` line with `message.usage` is one call. Streamed turns are written as several lines sharing one `message.id`; they are de-duplicated on that id across all files of a host. `agent-*.jsonl` files and `isSidechain` lines are labelled sub-agent. `cache_creation.ephemeral_5m_input_tokens` / `ephemeral_1h_input_tokens` split cache writes by TTL for pricing; transcripts without that breakdown are treated as 5-minute writes. Lines with model `<synthetic>` and zero usage are dropped.
- **Codex.** Each `event_msg` of type `token_count` with a non-null `info` is one call, using `last_token_usage`. Consecutive events whose cumulative `total_token_usage.total_tokens` did

not change are duplicates and dropped. Model and reasoning effort come from the most recent `turn_context`; the thread's `session_meta` supplies cwd, originator (CLI, Desktop, VS Code, exec) and version; `rate_limits.plan_type` on the same event records the billing plan. `input_tokens` includes cached tokens in Codex's schema, so uncached input = `input_tokens - cached_input_tokens`.

- Replay handling: the skill applies its gap-based filter to spawned/forked sessions. In this project dataset it skipped zero replay events. An independent comparison found no multiple session_meta replay blocks or cross-session copied usage fingerprints; every included scanner event matched the direct reference parser. Duplicate counter updates and zero-token state updates were excluded.
- **Copilot CLI.** One record per session from `session.shutdown.tokenDetails` (input, cache read, cache write, output); `assistant.message` events only carry output tokens and are used as a fallback when a session has no shutdown record.
- **opencode.** `message.data.tokens` and `cost` from `opencode.db`; the tool's own USD cost is used as-is.
- **OpenClaw.** `message.usage` on assistant messages; `<id>.checkpoint.<x>.jsonl` files replay the parent session and are de-duplicated on (message id, timestamp) across files.

2.2 Token taxonomy

Column	Meaning	Priced at
`input_uncached`	prompt tokens processed fresh	input rate
`cache_read`	prompt tokens served from the provider's prompt cache	cache-read rate
`cache_write`	cache-write tokens recorded by the harness; no cache-write tokens were logged in this dataset	cache-write rate for the TTL used
`output`	completion tokens, reasoning included	output rate
`reasoning`	the reasoning / thinking share of `output`	informational
`total`	sum of the first four	–

2.3 Pricing model

The standard API scenario applies the higher long-context rates to every call with more than 272,000 total input tokens, including cached input. Short-context rates below are in USD per million tokens. Batch/Flex discounts, Fast/Ultrafast premiums, regional premiums, subscription fees and provider tool charges are excluded. This is an API-equivalent estimate, not an invoice or observed spend. Exact model rates were verified on October 12, 2026 at https://developers.openai.com/api/docs/pricing.

Model	Input /M	Cache read /M	Cache write 5m /M	Cache write 1h /M	Output /M	Basis
`gpt-6-astra`	$10.00	$1.000	$12.50	$12.50	$50.00	published
`gpt-6.1-sol`	$2.00	$0.100	$2.50	$2.50	$10.00	published
`gpt-6-luna`	$0.10	$0.010	$0.12	$0.12	$0.50	published

2.4 Accounts and subscription attribution

Accounts are placeholders named after each host: credential files were not read, so plans and billing are unknown until `accounts.json` is filled in. Codex rollouts record the billing plan on every call (`rate_limits.plan_type`) but not the account, so calls are attributed to accounts by ordered rules in `accounts.json`; each rule carries its evidence and a confidence level. A subscription month is counted only when the account recorded at least one call in that month.

Unattributed calls. 16,189 calls (3.86B tokens, $3,320.68 API-equivalent) matched no rule in `accounts.json`. They are reported under the account `unattributed` with billing `unknown` rather than folded into any real account; add a rule to `accounts.json` to attribute them.

2.5 Validation against tool-maintained counters

Per-call usage was compared with an independent parser and the project-selected database counters. The two parsers agree on every included call and token field. No copied parent token records were detected in this dataset. One print-production counter was reset from its cumulative usage to 828,400 with a zero-token update; its per-call total is 168,060,439. An ongoing session also advanced beyond this fixed cutoff before the database snapshot was taken. These explain the two database disagreements; they do not establish replay inflation. The database snapshot is a different evidence class and time than this bounded per-call ledger.

Host	Threads compared	Within 2%	Within 10%	Rollout sum	SQLite sum	Ratio
local	189	98.9%	98.9%	3.86B	3.69B	1.045

Host	Thread	Source	Rollout	SQLite
local	`01a127d1-5a7d`	user	168.06M	828.40K
local	`01a13135-a9d7`	user	12.44M	14.83M
local	`01a127d6-4866`	user	364.91M	364.91M

3. Findings

F01 · Usage is concentrated in a small number of sessions within one project.

Confidence: High. Two working directories hold 77.4% of all tokens: `jevelopers-repository-execution-v1.1.0` (73.0% across 100 threads) and `shared-contracts-production-proposal` (4.4% across 1 threads). Within Codex, the largest thread holds 17.2% of tokens and the 18 largest threads (9.5% of 189) hold 75.1%; the largest single thread holds 17.2%. The two largest threads are `01a127d8…` in `jevelopers-repository-execution-v1.1.0` (gpt-6.1-sol, 2026-10-10 to 2026-10-12, 44 h, 662.61M) and `01a127d6…` in `jevelopers-repository-execution-v1.1.0` (gpt-6.1-sol, 2026-10-10 to 2026-10-11, 15 h, 364.91M).

Implication. Cost and capacity planning should be done per project and per thread, not per month; a few unattended multi-thread loops set the figures.

F02 · Prompt caching accounts for most input tokens.

Confidence: High. Codex re-sent a median of 193.26K prompt tokens per call (p90 454.64K, max 814.53K) while producing a median of 459 output tokens. 98.2% of prompt tokens were cache hits (Codex 98.2%). Prompt tokens per output token: Codex 239.

Implication. In the standard API scenario the cache avoided $25,116.87 (88.3%) relative to the uncached API-equivalent scenario. Context length, not output, is the cost driver.

F03 - Billing remains unverified

All 16,189 calls record plan_type=pro. Account identity, subscription fees, actual provider charges, and savings relative to a subscription are unverified. No credential files or invoices were read. The unattributed billing category means unknown, not zero spending.

F04 · Reasoning effort and sub-agent fan-out are material.

Confidence: High. Reasoning tokens are 53.5% of output (Codex 53.5%). Codex calls by configured effort: high 94.5%, max 4.6%, medium 0.5%, low 0.4%. Sub-agent share of tokens: Codex 14.4%.

Implication. Effort and subagent shares describe resource use. This study does not test alternative settings or hold task quality constant.

F05 · GPT-6.1 Sol leads token volume; Astra leads API-equivalent cost.

Confidence: High. gpt-6.1-sol carries 54.6% of all tokens (17.2% of API-equivalent cost). In 2026-10 the leading models by volume were gpt-6.1-sol (2.11B), gpt-6-astra (1.48B), gpt-6-luna (270.59M).

Implication. Blended cost per million output tokens: Codex $206.79; these are dominated by the context re-read, not the output rate.

F07 · All three model rate rows were verified; actual billing is unknown.

Confidence: Medium. 0.0% of the API-equivalent total ($0.00) rests on models priced by assumption. Halving or doubling every assumed rate moves the total to between $3,320.68 and $3,320.68.

Implication. The headline cost is robust to the pricing assumptions; it is not robust to the cache-read rate, which is where most of the discount comes from.

4. Results

4.1 By tool

Tool	Calls	Sessions	Uncached in	Cache read	Cache write	Output	Total	Hit rate	API-eq.	Cache saved
Codex	16,189	189	70.17M	3.77B	0	16.06M	3.86B	98.2%	$3,320.68	$25,116.87

4.1b Accounts and recorded plan

All 16,189 calls record plan_type=pro. Account identity, subscription fees, actual provider charges, and savings relative to a subscription are unverified. No credential files or invoices were read. The unattributed billing category means unknown, not zero spending.

Recorded plan	Calls	Tokens	Account identity	Actual charges
pro	16,189	3,859,621,041	Unverified	Unknown

4.2 By month

Figure 1 (tokens per month by tool) is rendered in the HTML version.

Figure 2 (API-equivalent cost per month) is rendered in the HTML version.

Month	Calls	Tokens	Output	Codex hit	API-eq.	Subscriptions	API/sub
2026-10	16,189	3.86B	16.06M	98.2%	$3,320.68	Unknown	–

Figure 3 (cache hit rate by month) is rendered in the HTML version.

4.3 Exact model totals

Model	Calls	Fresh input	Cached input	Output	Total	API-eq.
gpt-6.1-sol	8,077	35,054,202	2,062,375,680	9,931,497	2,107,361,379	$569.54
gpt-6-astra	7,366	31,732,108	1,445,109,632	4,823,400	1,481,665,140	$2,744.87
gpt-6-luna	746	3,382,081	265,908,992	1,303,449	270,594,522	$6.27

Daily usage in Eastern time

Eastern date	Calls	Total tokens	Output tokens	Standard API-equivalent
2026-10-10	3,147	812,579,981	3,811,824	$353.61
2026-10-11	6,965	1,592,919,234	7,170,258	$1,470.88
2026-10-12	6,077	1,454,121,826	5,076,264	$1,496.19

Context pricing tiers

Model	Calls at short rates	Calls at long rates	Tokens in long-context calls
gpt-6-astra	5,746	1,620	637,524,183
gpt-6-luna	266	480	227,382,836
gpt-6.1-sol	4,984	3,093	1,332,016,819

Short context is at most 272,000 input tokens. The higher tier is selected per call using fresh plus cached input. The complete rate sheet is in pricing.json.

Model summary

Model	Tool	Calls	Tokens	Share	Output	Hit rate	API-eq.
`gpt-6.1-sol`	Codex	8,077	2.11B	54.6%	9.93M	98.3%	$569.54
`gpt-6-astra`	Codex	7,366	1.48B	38.4%	4.82M	97.9%	$2,744.87
`gpt-6-luna`	Codex	746	270.59M	7.0%	1.30M	98.7%	$6.27

4.4 Distributions

Tool	Calls	Prompt p50	p90	p99	max	Output p50	p90	p99	max
Codex	16,189	193.26K	454.64K	696.86K	814.53K	459	2.42K	6.67K	117.26K

Tool	Sessions	Calls p50	p90	max	Tokens p50	p90	max	Span p50	p90	max
Codex	189	36	96	2,843	4.79M	25.22M	662.61M	0.2 h	1.4 h	44 h

4.5 Time of day

Calls by weekday and hour (America/New_York), Codex combined. Darker cells are more calls; the scale is square-root so unattended loops do not flatten everything else.

Figure 4 (weekday × hour heatmap) is rendered in the HTML version.

Busiest hours: 01:00 (1,437), 02:00 (1,239), 19:00 (1,127), 04:00 (1,071), 00:00 (1,048). Quietest: 14:00 (118), 15:00 (135), 10:00 (174).

4.6 Heaviest sessions

Thread	Main model	Calls	Tokens	Output	API-eq.
01a127d8-7f73	gpt-6.1-sol	2,843	662.61M	2.17M	$621.55
01a127d6-4866	gpt-6.1-sol	1,075	364.91M	1.57M	$101.77
01a127df-3ab3	gpt-6.1-sol	1,035	339.01M	2.09M	$111.07
01a129f4-461a	gpt-6-astra	958	312.12M	840.32K	$652.09
01a12e87-ec90	gpt-6.1-sol	783	278.68M	813.62K	$72.32
01a12e8b-b6e8	gpt-6-luna	437	169.88M	797.22K	$3.97
01a127d1-5a7d	gpt-6.1-sol	478	168.06M	809.41K	$49.50
01a12858-ad6c	gpt-6.1-sol	327	105.74M	409.63K	$27.46
01a12ddc-7345	gpt-6-luna	309	100.72M	506.23K	$2.30
01a128a7-26e3	gpt-6-astra	319	82.36M	378.43K	$171.09
01a12e88-7794	gpt-6.1-sol	250	54.08M	177.47K	$11.84
01a13003-000c	gpt-6-astra	158	50.98M	114.59K	$114.20
01a12821-dfd7	gpt-6.1-sol	168	42.37M	222.98K	$11.79
01a12f9f-cada	gpt-6-astra	191	41.40M	196.11K	$80.96
01a12e87-497c	gpt-6.1-sol	153	35.35M	96.27K	$7.68
01a1296d-b8be	gpt-6.1-sol	140	31.13M	159.88K	$8.11
01a12939-09d3	gpt-6.1-sol	126	29.82M	135.10K	$7.07
01a12932-098c	gpt-6-astra	86	28.40M	39.84K	$67.40
01a13054-737c	gpt-6-astra	128	25.99M	128.62K	$46.65

Thread	Main model	Calls	Tokens	Output	API-eq.
01a12cbe-99ac	gpt-6.1-sol	98	25.02M	145.59K	$7.01

Figure 5 (largest sessions) is rendered in the HTML version.

4.7 Working directories

Tool	Directory	Sessions	Calls	Tokens	API-eq.
Codex	jevelopers-repository-execution-v1.1.0	100	11,309	2.82B	$2,592.46
Codex	shared-contracts-production-proposal	1	437	169.88M	$3.97
Codex	production-output	1	327	105.74M	$27.46
Codex	shared-contracts-usage-proposal	1	309	100.72M	$2.30
Codex	runtime	4	262	54.28M	$14.80
Codex	canonical-adapters	3	221	50.87M	$13.76
Codex	production-paint-fix	1	126	29.82M	$7.07
Codex	production-final-review	1	86	28.40M	$67.40
Codex	shared-contracts-recovery	6	218	28.23M	$7.25
Codex	shared-contracts-production	2	122	26.93M	$7.27
Codex	composition	2	139	23.43M	$5.23
Codex	durability	3	151	20.75M	$5.00
Codex	storage-verification	1	81	16.95M	$4.85
Codex	shared-contracts-review-10d9710	1	65	16.75M	$36.07
Codex	review	1	65	16.26M	$34.32
Codex	shared-contracts-review-b47defa	1	62	15.38M	$32.98
Codex	shared-contracts-review-c72a939	1	63	14.94M	$31.25
Codex	shared-contracts-review-795608e	1	55	12.81M	$27.16
Codex	shared-contracts-review-9820f68	1	56	12.36M	$25.94
Codex	exporter-conformance	1	74	11.50M	$2.42

4.8 Entry points, effort and versions

Tool	Entry point / originator	Calls	Tokens	API-eq.
Codex	Codex Desktop	16,189	3.86B	$3,320.68

Tool	Effort setting	Calls	Output	Reasoning share	API-eq.
Codex	high	15,304	14.69M	52.8%	$3,287.28
Codex	max	746	1.30M	63.8%	$6.27

Tool	Effort setting	Calls	Output	Reasoning share	API-eq.
Codex	medium	82	37.30K	9.7%	$15.39
Codex	low	57	23.30K	2.9%	$11.73

Client versions seen: **Codex** 0.162.0-alpha.2 (16,189).

5. Uncertainty and limitations

- **Retention.** Claude Code transcripts older than the retention window are gone (F06). Only retained local project sessions were measured; missing or deleted logs cannot be reconstructed. Copilot CLI keeps only recent sessions.
- **Replay filter.** The gap rule that strips replayed history from forked Codex rollouts is a heuristic. A genuine first response arriving within 2 s of the end of the replay burst would be dropped; a pause of more than 2 s inside the burst would leave the rest of the parent's history counted. The per-thread comparison in 2.5 bounds the effect.
- **Prices are list prices at one date.** Provider rates change; everything is priced at the sheet in `pricing.json`. Historical months are approximations, not invoices.
- **Tokenizer.** Token counts across model generations are not directly comparable as measures of text volume.
- **Subscription counterfactual.** The API-equivalent figure assumes identical traffic on a pay-as-you-go key. In practice each subscription's rate limits shaped the traffic.
- **Account attribution.** Codex records the billing plan per call but not the account, so accounts are assigned by rules whose evidence and confidence are listed in 2.4; medium- and low-confidence rules are the ones to challenge first.
- **Timestamps.** Tool timestamps are UTC at write time; time-of-day analysis converts to America/New_York. Sessions spanning DST changes are not adjusted.
- The inclusive cutoff is 2026-10-12T18:31:50.391Z (October 12, 2026, 2:31:50 p.m. Eastern). It matches the earlier 3,859,621,041-token answer. Per-day tables use America/New_York.
- The 192 selected session files contain 189 sessions with usage and 16,189 nonzero calls. Three launch-probe sessions have no recorded tokens.
- An independent parser matched every included call. No cross-session copied usage fingerprints or multiple session_meta replay blocks were present. The skill replay filter removed zero events; its replay fixtures passed.
- Print-production session 01a127d1-5a7d-7b43-8e0f-179dddbf9814 contains 168,060,439 per-call tokens, while its database counter was reset to 828,400 with a zero-token update. Use the per-call total. Database counters for ongoing sessions may also exceed this fixed cutoff.
- Costs are a standard-tier API-equivalent scenario using verified list rates and per-call long-context premiums, not actual charges or an invoice. The actual API processing tier and regional premiums were not recorded by this scanner.
- Account identity, actual subscription fees, subscription savings and allocation across billing accounts are unknown. Call-level plan labels are retained as recorded; they do not verify an account or invoice.
- The skill labels native collaboration children as subagents. Separately launched CLI implementation/review workers can appear in the main category.
- Model volume does not establish task quality, completion, developer hours saved or project business value.

6. Reproducibility

Generated using AI Usage Ledger skill v1.6.0 at commit e9d5e225a802d8eddcc7e1a2fc72cded6db1437e. project-report.mjs scopes the skill scanner to selected project sessions, freezes the time interval, removes zero-token updates, verifies each call independently, and normalizes daily dates to Eastern time. Source logs remain unchanged.

``` node project-report.mjs python3 skill-source/scripts/run_pipeline.py --manifest manifest.json --only report,analyze,build node finalize-report.mjs ```

File	SHA-256
compile_ai_logs.py	`453143bb0a944e0063ab4c2375f7c14d737cad44595e141b3f5dd7e879fe0f68`
analyze_events.py	`290d4028865816190c4f8d10ac2164b339584bad01c944d240ec6dfe52350b3d`
build_report.py	`02509ccffb8a5f3746a3fc24063593ec2fa2f7196ce87627f6aeb6c3c65f2236`
pricing.json	`276694c2a15d418ed8f3fe81d37528fa4108e542b09d18e7b6a4b2120c5166be`
summary.json	`c4ee03f9aa77dc92bbc7d3adf2887040ba392858412d3acac012b435cce290a0`
analysis.json	`b33e7deee7b0c0a856f53117638e043aaa381579d47febd78f9eb84ce27780b5`

The event-level dataset (`compiled/all_events.csv`, 16,189 rows, one per model call with every token class, the pricing row applied, the account assigned and the source file path) is the primary artefact; every table above is an aggregation of it.

Appendix A. Glossary

Term	Definition
Call	One model request whose usage the tool recorded: an assistant message (Claude Code, OpenClaw, opencode), a `token_count` event (Codex), or a session (Copilot CLI).
Session / thread	One transcript file (Claude Code), one rollout file (Codex), one OpenClaw session file, one opencode session, one Copilot session.
Sub-agent	A Codex thread spawned by another thread (`source.subagent`) or a Claude Code `agent-*.jsonl` transcript.
Cache hit rate	`cache_read ÷ (input_uncached + cache_read + cache_write)`.
API-equivalent cost	Tokens × list price per token class, from `pricing.json`.
Cache saving	`cost_if_uncached − api_cost`, where `cost_if_uncached` prices every prompt token at the input rate.
Account	A login identified from a credential file (or a per-host placeholder when credential files were not read); assigned to calls by the rules in accounts.json.
Usage-based	A call billed per token to a prepaid or metered org rather than covered by a subscription.

Appendix B. Package contents

- `USAGE_REPORT.md` — this document.
```

- `report.html` — the same document with the figures rendered.
- `ledger.json` — machine-readable totals, monthly and model tables, top sessions, validation results, sensitivity, accounts, source inventory and the pricing sheet used (scan roots and working directories reduced to their last path component; the report configuration omitted).
- `pricing.json` — the rate sheet applied.
- `SENSITIVITY.md` — what in this package names hosts, directories, accounts or paths, and how to check or anonymise it before sharing.
- `SHA256SUMS` — binds the files above.